

ATHENA Championship System

Architecture Documentation

1. System Overview

The Athena Championship Platform is a microservice-based system designed to manage multi-sport championships.

It supports user management, championship management, event tracking, scoring, ranking, and secure authentication.

2. High-Level Architecture

Architecture follows microservices principles with clear service boundaries.

Core services include user-service, championship-service, event-service, scoring-service, ranking-service.

3. Microservices Responsibilities

User Service: Authentication, authorization, role management.

Championship Service: Championship lifecycle management.

Event Service: Match scheduling and state handling.

Scoring Service: Discipline-specific scoring strategies.

Ranking Service: Leaderboards and ranking calculations.

4. Security Architecture

JWT-based stateless authentication.

Role-based and permission-based authorization using method-level security.

Bearer token passed in Authorization header.

5. Database Architecture

Database per microservice pattern.

No cross-service joins.

Future support for Flyway/Liquibase migrations.

6. Communication

Synchronous REST communication between services.

Future support for asynchronous messaging (Kafka/RabbitMQ).

7. Deployment

Each service runs in its own Docker container.

Environment profiles: dev, test, prod.

8. Testing Strategy

Controller tests using MockMvc.

Service tests using Mockito.

Repository tests using DataJpaTest.

Integration tests using SpringBootTest.

Code coverage using JaCoCo.

9. Scalability & Observability

Horizontal scaling support.

Stateless services.

Future centralized logging and monitoring integration.